Model

1 Model

1.1 set up

Consider a metropolitan area consisting of a finite set of locations $S = \{1, 2, \ldots, N\}$. These locations are partitioned into a set of municipalities $g \in G$, such that

$$S = \bigcup_{g \in G} S^g, \qquad S^g \cap S^{g'} = \emptyset \quad \text{for } g \neq g'.$$

Each location $i \in S$ is endowed with a fixed amount of land $\bar{H}_i$, which can be allocated between residential and commercial use:

$$H_i^R + H_i^F \leq \bar{H}_i.$$

Land use is further subject to zoning regulations. Let Z_i^R and Z_i^F denote the residential and commercial zoning restrictions at location i. These regulations impose upper bounds on feasible development:

$$H_i^R \leq \bar{H}_i^R(Z_i), \qquad H_i^F \leq \bar{H}_i^F(Z_i),$$

where stricter zoning lowers the feasible density of development.

Locations are connected through a transportation network. Following **Fajgelbaum2020-yv**, the generalized commuting cost is denoted by τ_{ij}, which depends on transportation infrastructure and congestion:

$$\tau_{ij} = \tau_{ij}(I_{ij}, \mathbf{n}),$$

where I_{ij} denotes the transportation investment between locations i and j, and $\mathbf{n}$ collects equilibrium traffic and transit usage across routes, which generate congestion. The commuting cost is increasing in the quantity of commuters on the route

$$\frac{\partial \tau_{ij}}{\partial n_{ij}} \geq 0, \tag{1}$$

and I_{ij} leads to a reduction in the commuting cost

$$\frac{\partial \tau_{ij}}{\partial I_{ij}} \leq 0. \tag{2}$$

Following **Fajgelbaum2020-yv**, I will parametrize the commuting cost function as follows:

$$\tau_{jk}(\mathbf{n}, I) = \delta_{jk}^{\tau} \frac{(\mathbf{n})^{\rho}}{\mathbf{I}^{\gamma}} \tag{3}$$

There are three types of agents in the model: households, firms, and governments. Households choose where to live, where to work, and how to commute. Competitive firms choose labor and commercial land to produce a freely tradable good. Government decisions are made at two levels. Municipal governments choose local zoning regulations, while a metropolitan transportation authority chooses transportation investment for the urban area as a whole.

1.2 Household

For a household ω located at i and working at j, the utility is given by

$$u_{\omega ij} = \frac{B_i}{d_{ij}}(\frac{C_{ij}}{\alpha})^{\alpha}(\frac{H_{ij}}{1-\alpha})^{1-\alpha}v(\omega ij)$$

where C_{ij} and H_{ij} are the consumption of the composite good and housing, respectively, and $v(\omega ij)$ captures idiosyncratic preferences for location and job pair (i, j). The term B_i captures the local amenities at residence location i, while d_{ij} captures the spatial friction of commuting from i to j. The household faces a budget constraint:

$$C_{ij} + q_i H_{ij} \leq w_j$$

Solved: $V_{ij\omega} = \frac{B_i w_j q_i^{\alpha-1}}{d_{ij}} v_{ij\omega}$ and $F(v_{ij\omega}) = exp(-T_i E_j v_{ij\omega}^{-\epsilon})$

Such that

$$\pi_{ij} = \frac{T_i E_j (\frac{B_i w_j q_i^{\alpha-1}}{d_{ij}})^{\epsilon}}{\sum\sum T_r E_s (\frac{B_r w_s q_r^{\alpha-1}}{d_{rs}})^{\epsilon}} = \frac{\Phi_{ij}}{\Phi} \tag{4}$$

Given equilibrium condition

$$E(V_{ij\omega}) = \Gamma(\sum\sum T_r E_s (\frac{B_r w_s q_r^{\alpha-1}}{d_{rs}})^{\epsilon})^{1/\epsilon} = U \tag{5}$$

1.3 Firm

A firm located at j produces output using labor and commercial land from location j according to the production function

$$y_j = A_j L_{mj}^{\beta}(H_j^F)^{1-\beta}$$

where H_j denotes the commercial land available at location j. L_{mj} denotes the labor demand from location j. For now I ignore the agglomeration effect and assume that the productivity A_j is location-specific and exogenous.

1.4 Local Municipal

For each municipality g, the local government chooses zoning regulations to maximize the welfare of its incumbent residents.

1.5 Optimal Transportation network

1.6 Social planner

2 Model 2

In this section, I try to build a model with two time periods to see if it help me capture the mechanism of zoning regulations better. In the first period There are existing incumbents who own the land and decide the zoning regulations for second period, wheras the government choose the transportation investment given the whole zoning regulations in each municipality. In the second period, the new entrants choose where to live and work given the zoning regulations and transportation network determined in the first period.

2.1 set up

Consider a metropolitan area consisting of a finite set of locations $S = \{1, 2, \dots, N\}$. These locations are partitioned into a set of municipalities $g \in G$, such that

$$S = \bigcup_{g \in G} S^g, \qquad S^g \cap S^{g'} = \emptyset \quad \text{for } g \neq g'.$$

Each location $i \in S$ is endowed with a fixed amount of land $\bar{H}_i$. In the first period, each location i is occupied by incumbents who own the land and decides the zoning regulations for the second period.